Skills & Connectors: Teach AI Once, Connect It to Your Apps

Foundations (Everyone), Chapter 6/10 — Revision & Viva-Prep Cheatsheet (English + Roman Urdu)
Source: agentfactory.panaversity.org/docs/skills-connectors-crash-course

EN: A Skill teaches AI a task once (a folder + SKILL.md, loaded only when needed). A Connector gives it safe access to your apps (MCP, inheriting your permissions). Kitchen analogy: Connectors are the kitchen, Skills are the recipe cards.
Roman Urdu: Skill AI ko ek task ek baar sikhati hai (folder + SKILL.md, zaroorat par load). Connector safe access deta hai apps tak (MCP, aapki permissions inherit karta). Kitchen analogy: Connectors kitchen hain, Skills recipe cards.

1. Two Upgrades

- **Skill** = a folder + SKILL.md, progressive disclosure (only loads when relevant). **Connector** = MCP -- inherits YOUR permissions, is read-only or write, scoped per-conversation.
- Skill vs Connector vs Project vs Custom Instructions: each solves a different standing-context problem. Automatic firing vs an explicit slash command.

2. Using What Exists

- Built-in Word/PPT/Excel/PDF skills live behind the 'Code execution' switch. Connecting an app takes about 1 minute, enabled per-chat.

3. Building Your Own

- skill-creator = fastest path. **SKILL.md anatomy:** frontmatter is always-loaded, use kebab-case naming.
- The **description field is the whole game** -- it decides when the skill fires; include negative triggers too. Test-then-iterate before sharing.

4. Five Places + Safety

- Agent Skills is an open standard (agentskills.io, Agentic AI Foundation) -- Skills ARE portable across 5 surfaces; ChatGPT Custom GPTs / Gemini Gems are NOT (vendor-locked).
- **2 real risks:** malicious skills (prompt injection + data exfiltration), over-broad connector write access. 7-item safe-use checklist. Scope is a Connector concern, not a Skill concern.

Key Terms Cheatsheet

One line each — English + Roman Urdu.

Term	English	Roman Urdu
Skill	A folder + SKILL.md that teaches AI a task, loaded only when relevant.	Folder + SKILL.md jo AI ko task sikhati, zaroorat par load hoti.
Connector	MCP-based access to an app, inheriting your own permissions.	MCP-based app access, aapki apni permissions inherit karta hai.
Progressive disclosure	A skill's content loads only when actually needed, not all at once.	Skill ka content tabhi load hota jab zaroorat ho, sab ek sath nahi.
Description field (SKILL.md)	Decides when a skill fires -- the single most important field.	Decide karta skill kab fire hogi -- sabse zaroori field.
Portable vs vendor-locked	Agent Skills (open standard) are portable; Custom GPTs/Gems are not.	Agent Skills (open standard) portable hain; Custom GPTs/Gems nahi.

Quick Self-Check

Question	Answer
What's the difference between a Skill and a Connector, in one line?	A Skill teaches a task once; a Connector grants safe access to an app.
Why is the SKILL.md description field 'the whole game'?	It's what decides when the skill actually fires -- including its negative triggers.
Why are Agent Skills portable but ChatGPT Custom GPTs are not?	Skills follow an open standard (agentskills.io); Custom GPTs/Gems are vendor-locked to one platform.
What are the two real safety risks this chapter names?	Malicious skills (prompt injection + data exfiltration) and over-broad connector write access.

Where This Leads / Sources

Leads to: 5 practice projects (build first skill, connect app read-only, wire together, make portable, audit a skill). Full 29-Q quiz in 07-test-your-understanding.md.

Source: docs/skills-connectors-crash-course/ (this repo), content originally via Zia Tutor AI connector.